

# Elasticsearch Installation on Vultr Cloud – SOC-ELK Server

Complete Step-by-Step Visual Guide

From VPC creation to a running Elasticsearch 9.3.3 instance

**Server name:** SOC-ELK

**Cloud provider:** Vultr

**OS:** Ubuntu 22.04 LTS x64

**Plan:** 4 vCPU / 16 GB RAM / 80 GB NVMe

**Elasticsearch:** 9.3.3 (DEB package)

**VPC:** Day1-MYDFIR-SOC (172.31.0.0/24)

This guide walks you through creating a VPC, deploying an Ubuntu server, installing Elasticsearch, configuring the network, setting a firewall, and verifying the service.

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# 1. Prerequisites

Before you begin, ensure you have:

• A Vultr account with sufficient credit or payment method.

• Access to the Vultr Customer Portal (my.vultr.com).

• An SSH client (PowerShell, Terminal, PuTTY, etc.).

• Basic familiarity with Linux command line.

**Note:** The screenshots show a lab named Day1-MYDFIR-SOC. You can use any naming convention that fits your environment.

# 2. Create VPC Network

Log in to the Vultr Customer Portal at **https://my.vultr.com**.

In the left sidebar click **Products**, then expand **Network** and select **VPC Networks**.

Click the blue **Create VPC Network +** button.

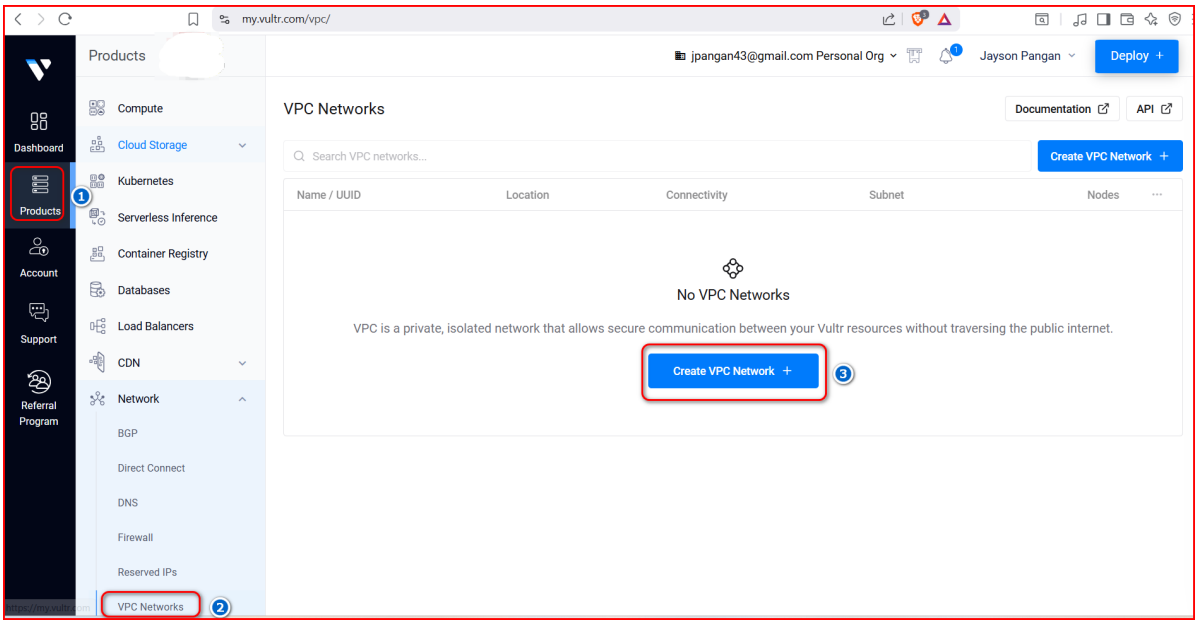



*Figure 1 – Vultr Products → Network → VPC Networks*

## 3. Configure VPC Name & Connectivity

Enter a descriptive name, e.g. **Day1-MYDFIR-SOC**.

Choose **Instances with Public IP** (the free option). This allows the server to have a public IP while still belonging to the private VPC.

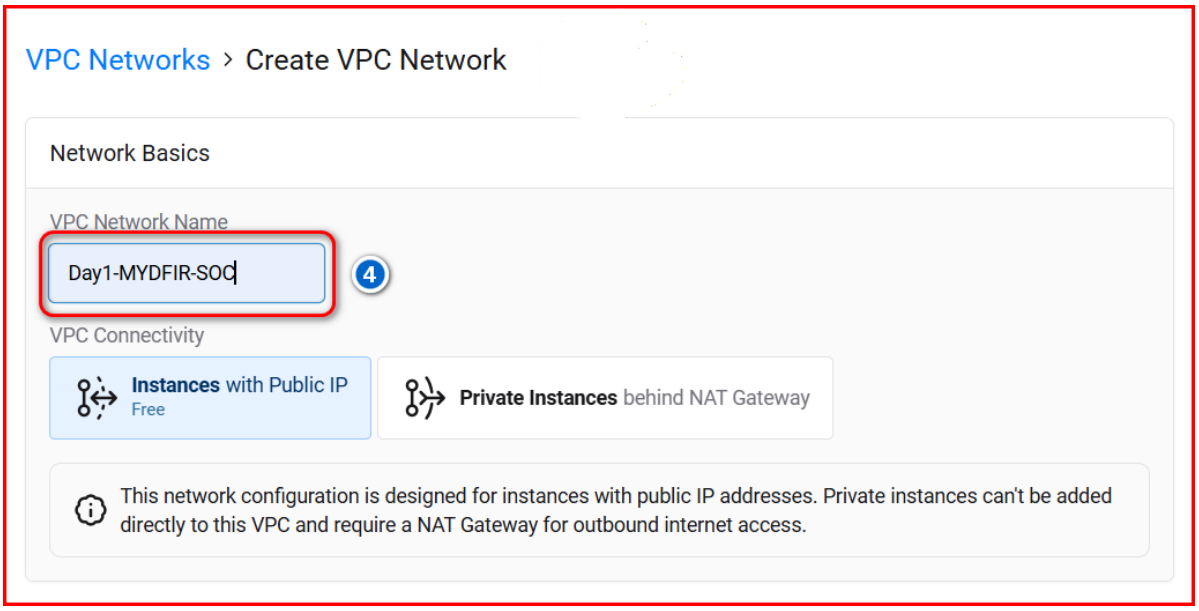



*Figure 2 – VPC Network Name and Connectivity type*

# 4. Select Location

Under Location, select the same region where you plan to deploy the server (here: **New York, US**).

Keeping the VPC and the instance in the same region is required for attachment.

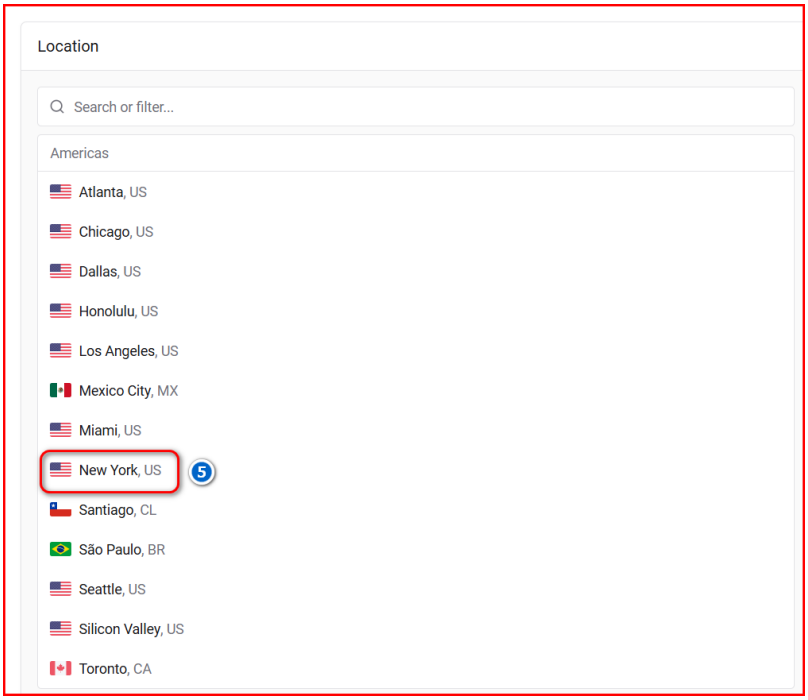



*Figure 3 – Choose location (New York, US)*

## 5. Set Custom IP Range

Enable **Custom IP Range**.

Set IPv4 Network Address to **172.31.0.0** and Network Prefix to **24** (i.e. 172.31.0.0/24).

This private subnet will be used by the SOC-ELK server.

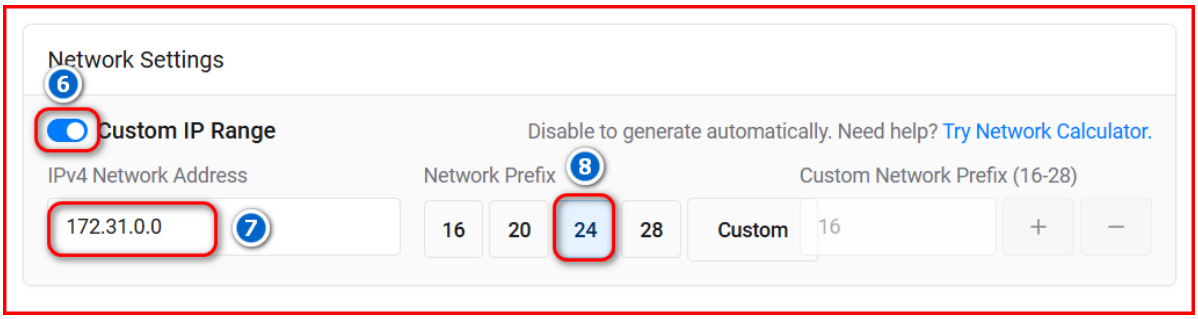



*Figure 4 – Custom IP range 172.31.0.0/24*

## 6. Create the VPC

Review the summary (Name, Connectivity = Public IP, Location = New York).

Click **Create VPC Network +**.

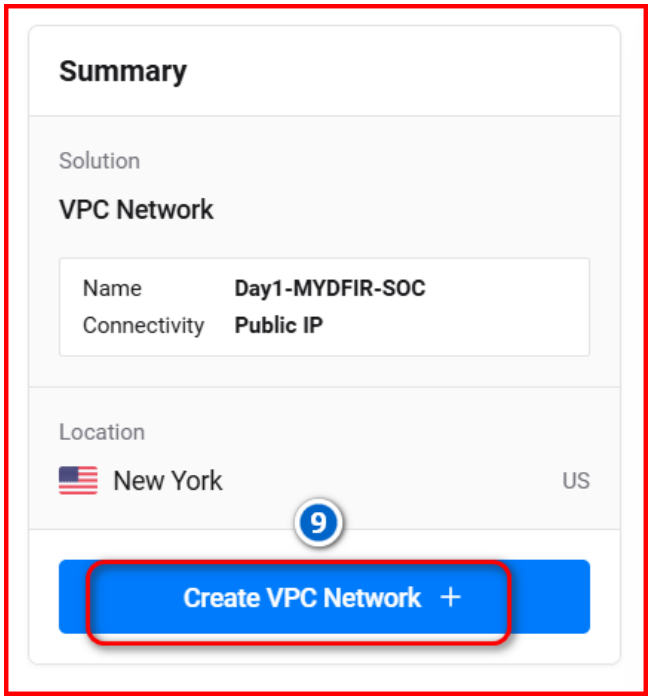



*Figure 5 – Summary and Create VPC Network*

# 7. Confirm VPC Created

You should now see the new VPC in the list with its UUID, location, connectivity type and subnet.

Copy the UUID if you need it later; the name is usually enough for selection.

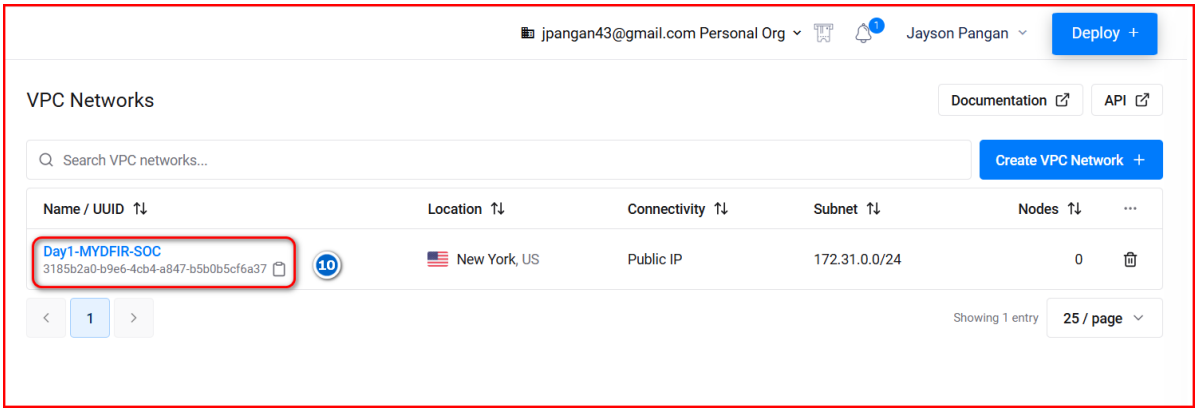



*Figure 6 – VPC listed with subnet 172.31.0.0/24*

# 8. Deploy Server – Choose Type & Location

Click **Deploy +** (top right) or go to Products → Compute.

Select **Dedicated CPU**.

Under Americas choose **New York (NJ)** so it matches the VPC region.

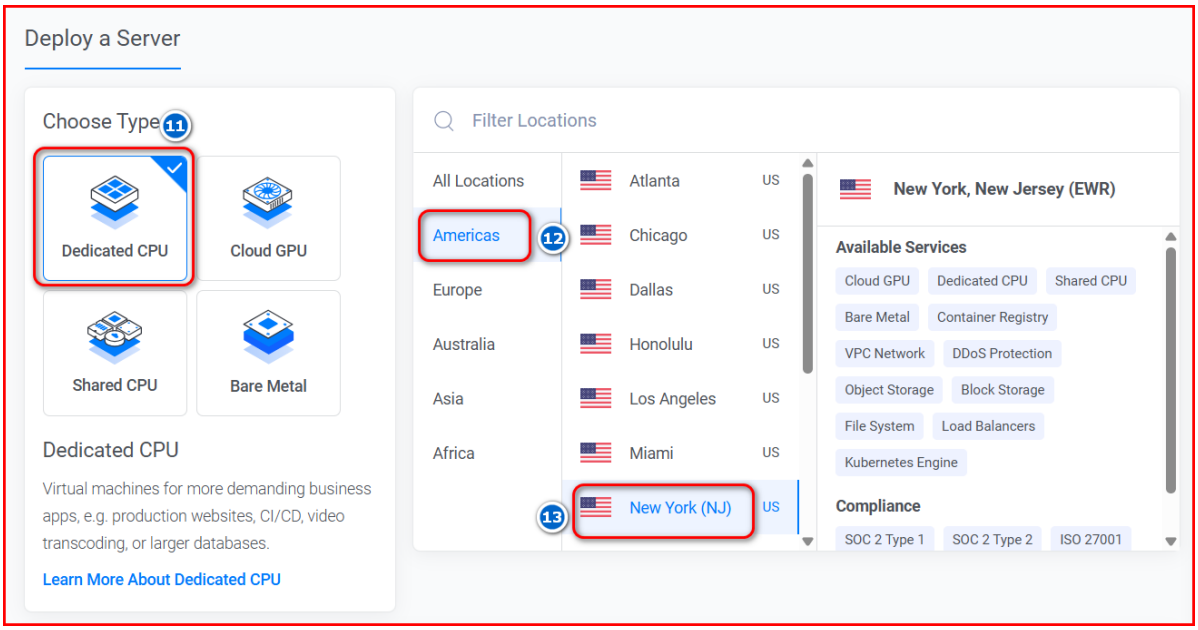



*Figure 7 – Deploy a Server: Dedicated CPU + New York (NJ)*

# 9. Select Server Plan

In the General Purpose category select the plan **voc-g-4c-16gb-80s** (4 vCPUs, 16 GB RAM, 80 GB NVMe).

Leave Automatic Backups disabled for this lab (or enable them if desired).

Click **Configure**.

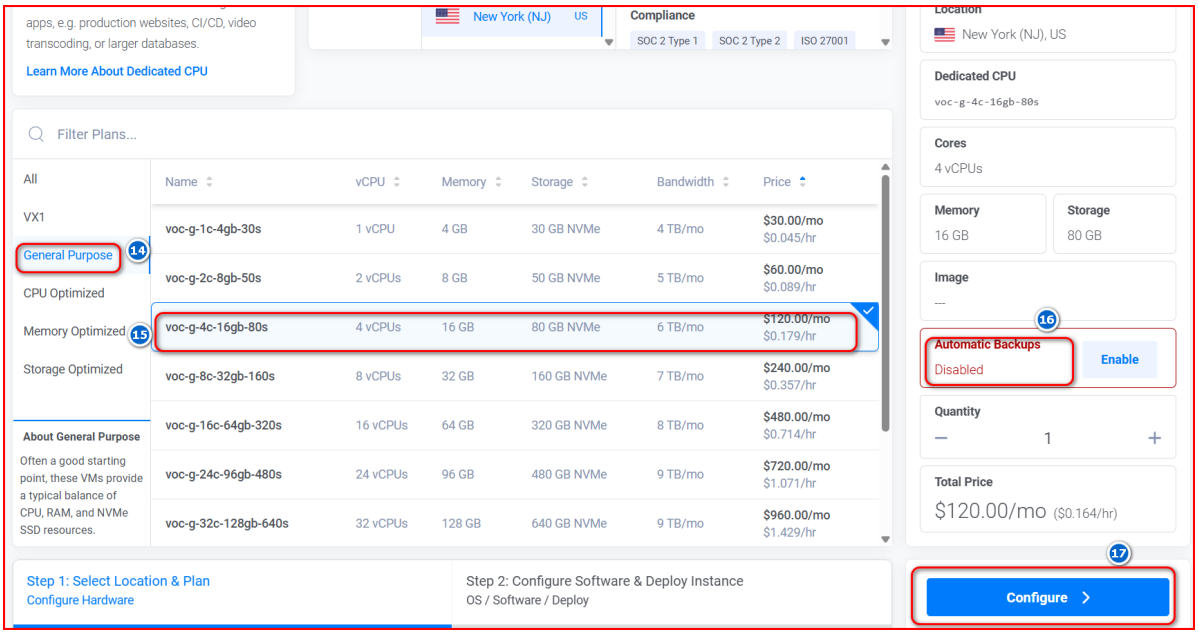



*Figure 8 – Plan voc-g-4c-16gb-80s (4 vCPU / 16 GB / 80 GB)*

# 10. Choose Operating System (Ubuntu 22.04)

On the Operating System tab select **Ubuntu** → **22.04 x64**.

This is a long-term support release well supported by Elasticsearch.

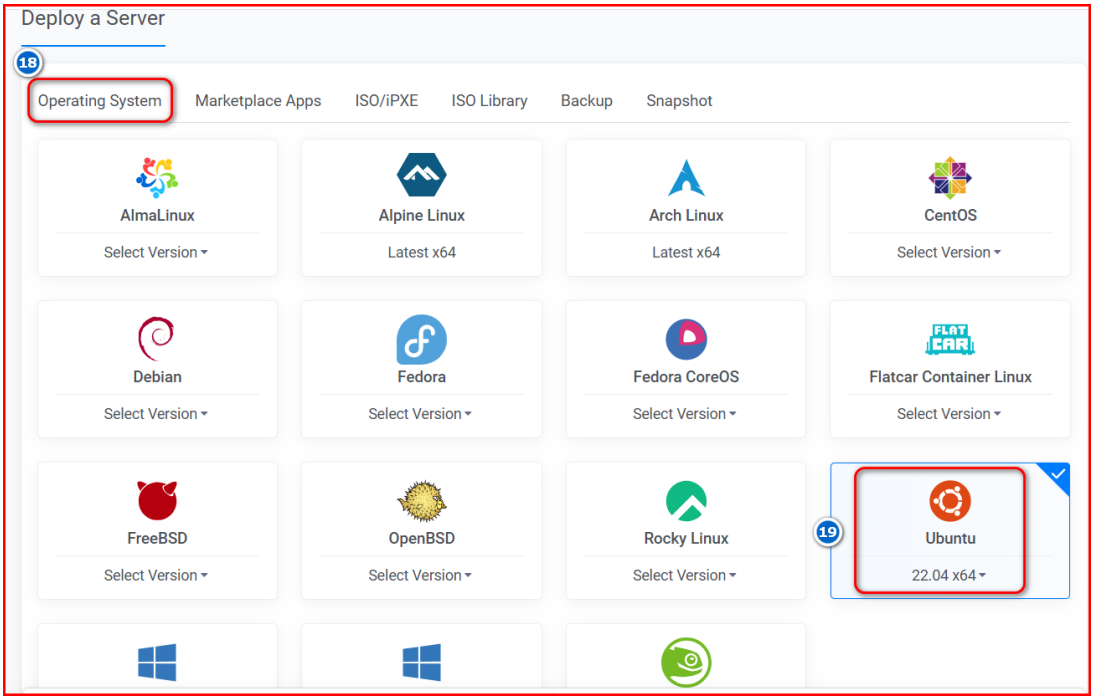



*Figure 9 – Operating System: Ubuntu 22.04 x64*

# 11. Set Hostname and Public IP

Set both Hostname and Label to **SOC-ELK**.

Keep **Instance(s) with Public IP** and enable **Public IPv4**.

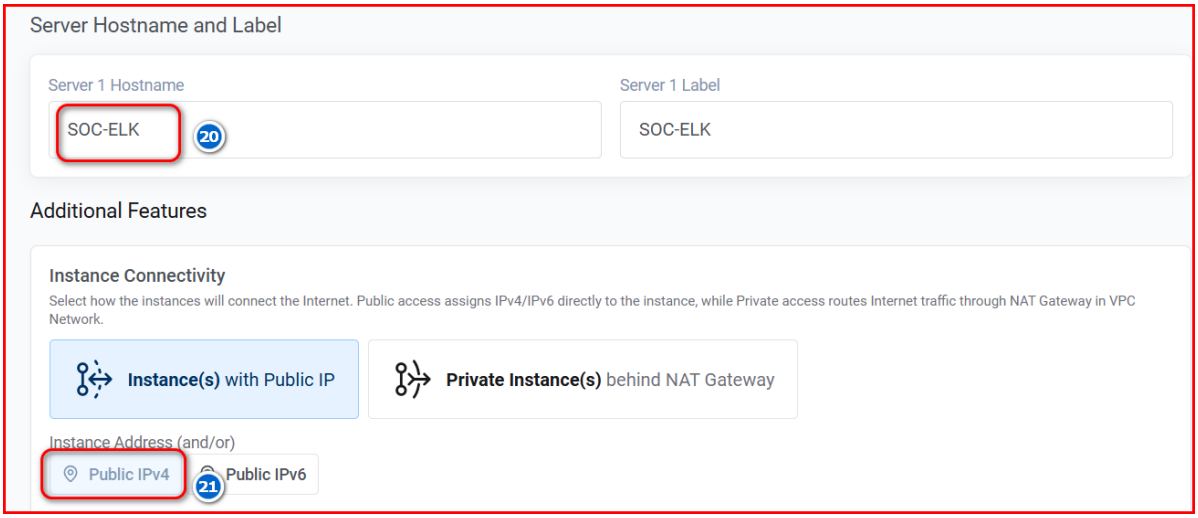



*Figure 10 – Hostname SOC-ELK and Public IPv4*

# 12. Attach VPC and Deploy

Enable the VPC Network toggle and check the box for **Day1-MYDFIR-SOC**.

Review the summary (Ubuntu 22.04, 16 GB, Public IPv4, VPC attached).

Click **Deploy**. The instance will take a few minutes to finish provisioning.

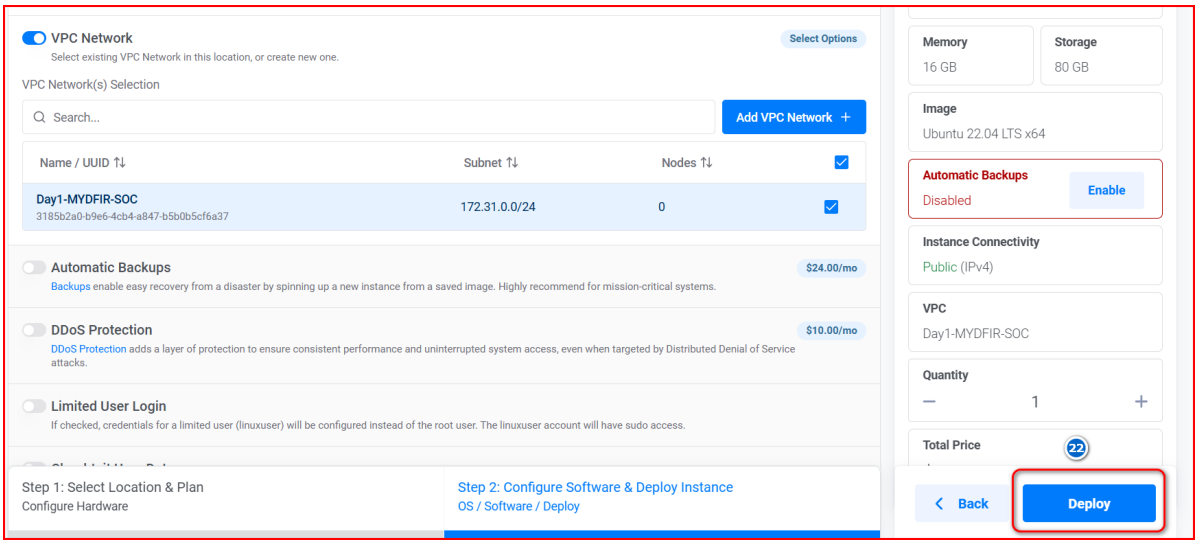



*Figure 11 – Select VPC Day1-MYDFIR-SOC and Deploy*

# 13. Server Ready – Copy IP & SSH

Once the server shows as running, note the **public IP address** and the root password (or use SSH key if configured).

From your local machine open a terminal / PowerShell and connect:

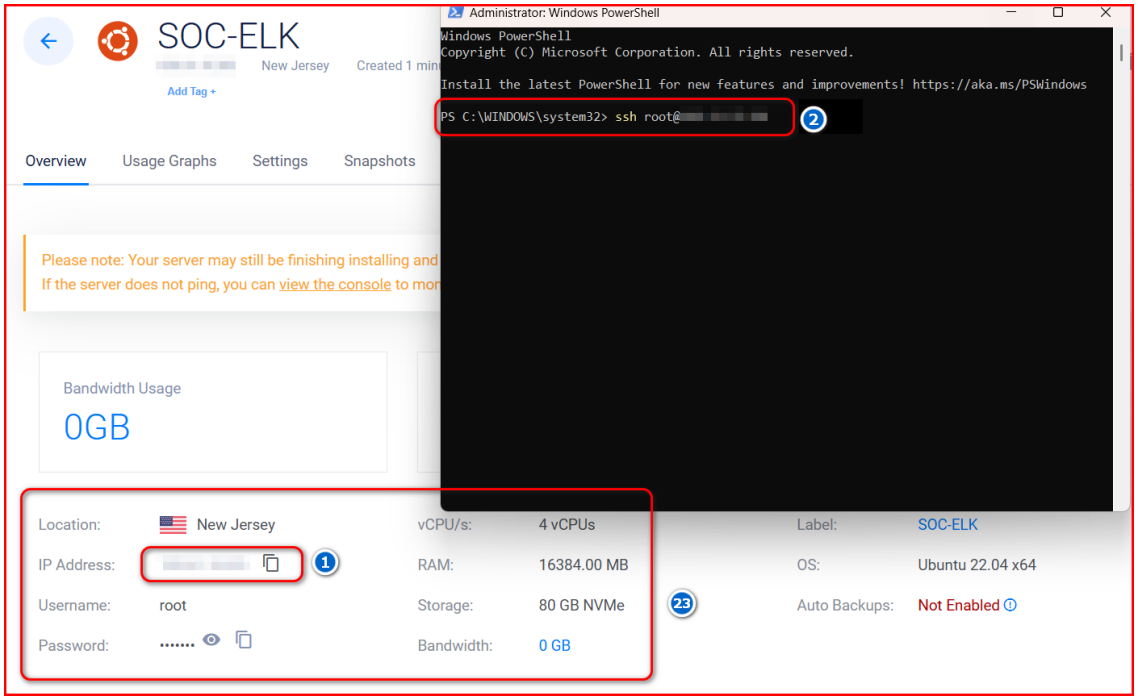



*Figure 12 – Server overview with public IP and SSH command*

```
ssh root@<PUBLIC-IP>
```

## 14. First Login as root

Accept the host key if prompted. You should land at the root prompt:

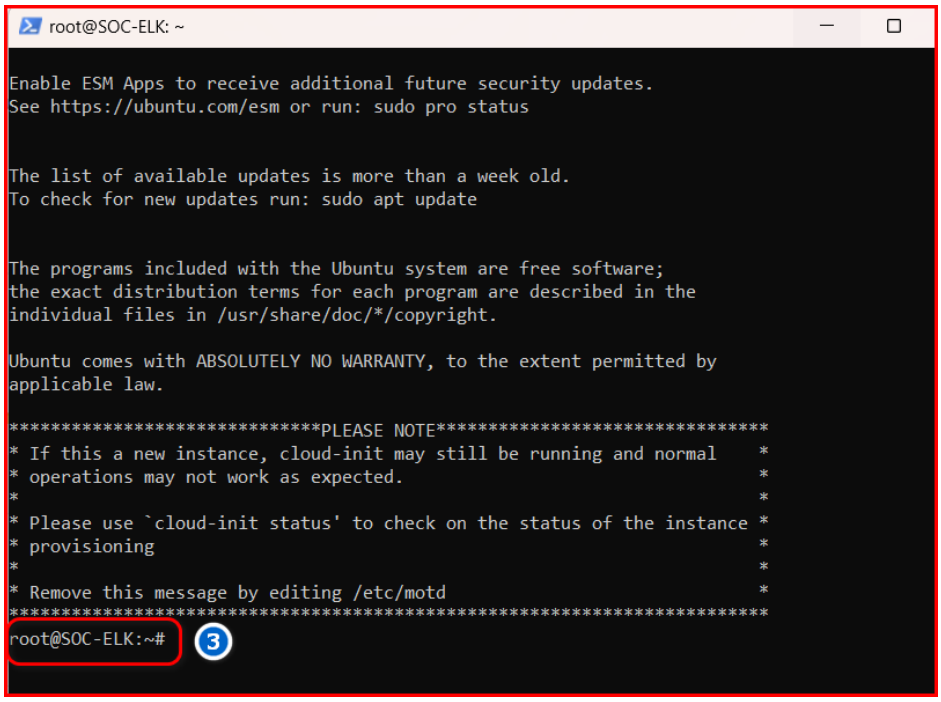



*Figure 13 – Successful SSH login as root@SOC-ELK*

```
root@SOC-ELK:~#
```

# 15. Update the System

Bring the system fully up to date before installing any software:

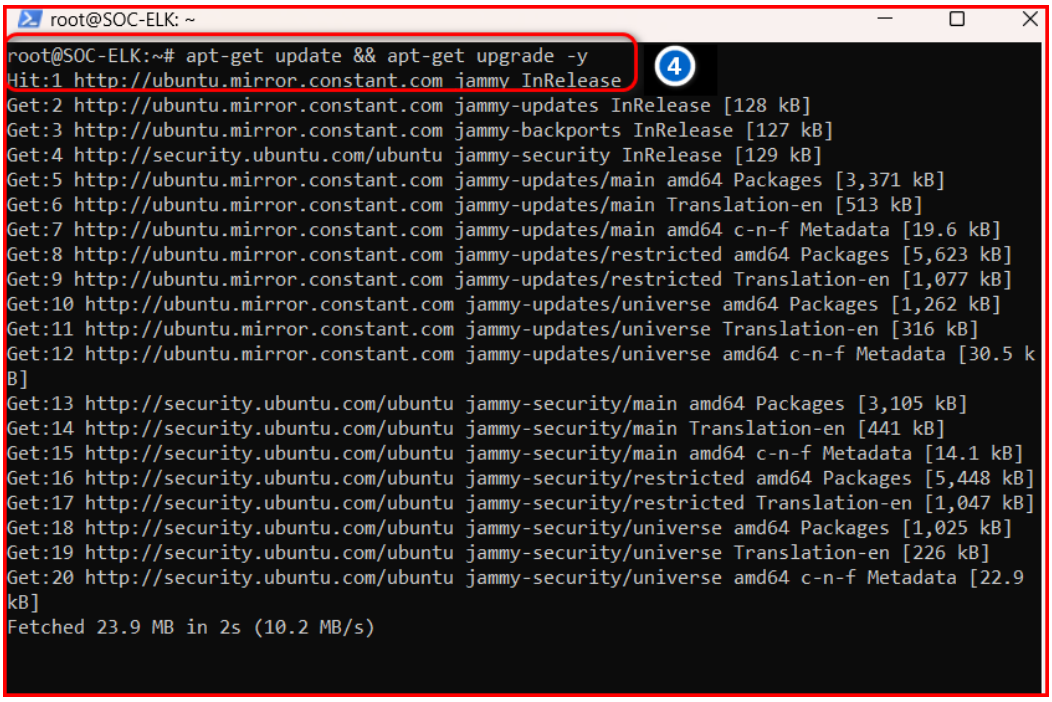



*Figure 14 – apt-get update && apt-get upgrade -y*

```
apt-get update && apt-get upgrade -y
```

Wait until the upgrade finishes (may take several minutes).

## 16. Download Elasticsearch 9.3.3

Download the official Debian package from Elastic's artifact repository:

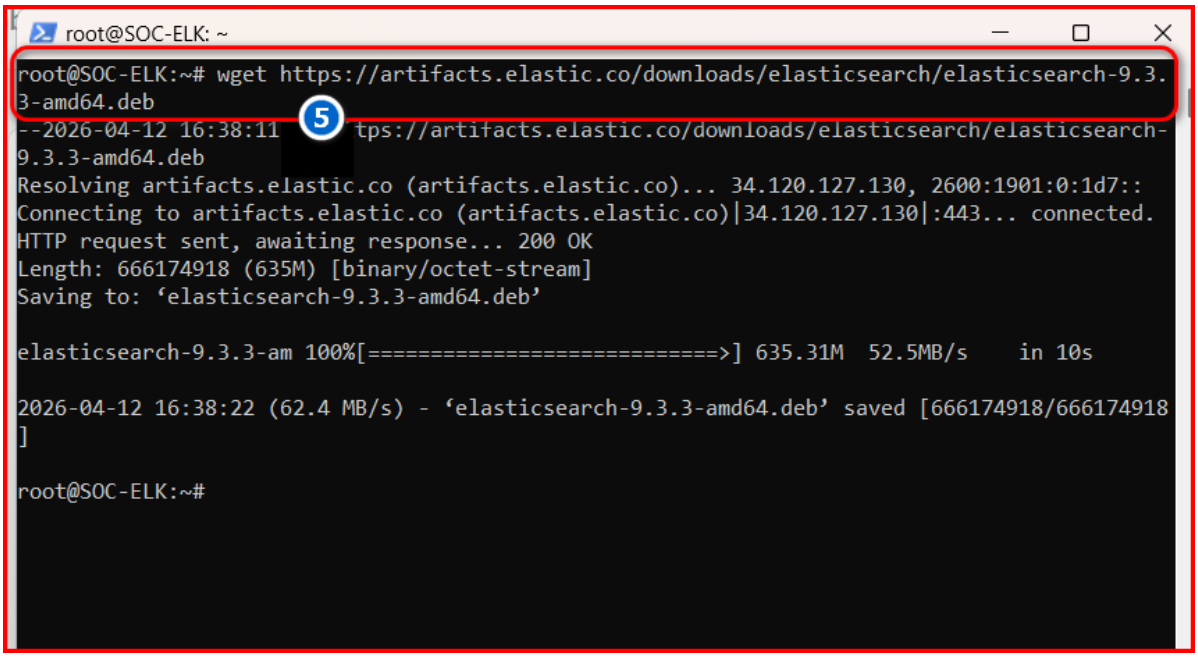



*Figure 15 – wget Elasticsearch 9.3.3 DEB package*

```
wget https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.3.3-amd64.deb
```

The package is approximately 635 MB.

## 17. Install the DEB Package

Install the downloaded package:

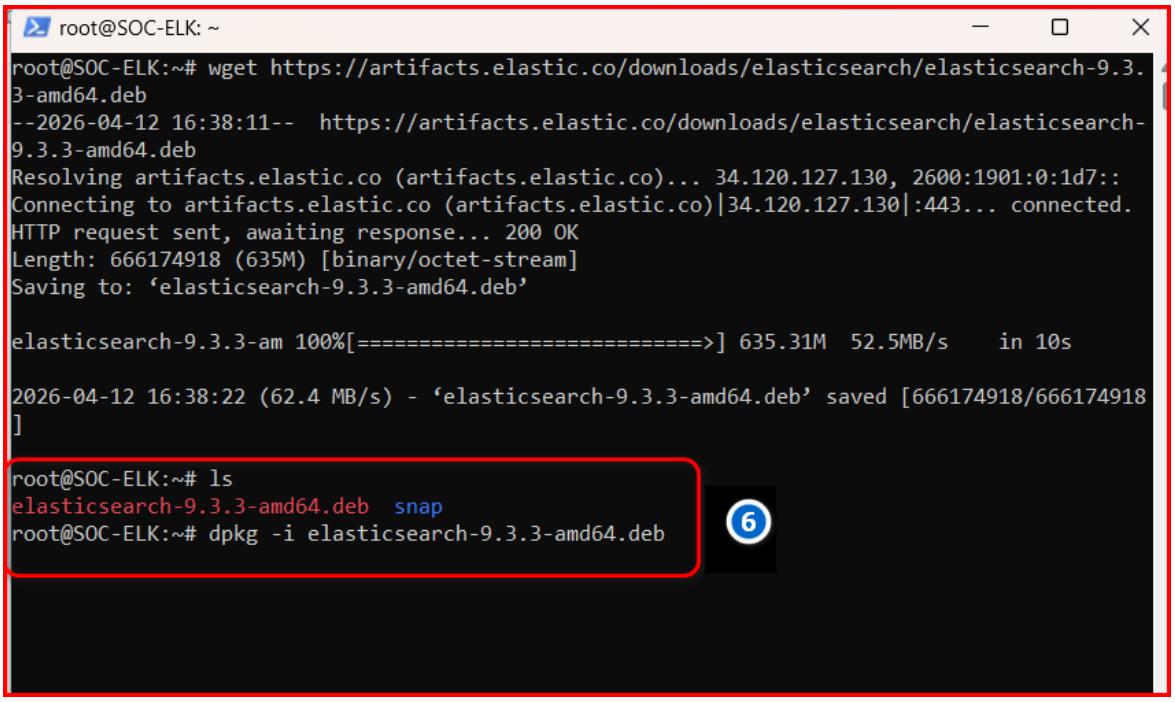



*Figure 16 – dpkg -i elasticsearch-9.3.3-amd64.deb*

```
dpkg -i elasticsearch-9.3.3-amd64.deb
```

# 18. Note the elastic Superuser Password

During installation Elasticsearch generates a random password for the built-in **elastic** superuser.

**Copy and store this password securely** – you will need it to access the cluster.

You can later reset it with:

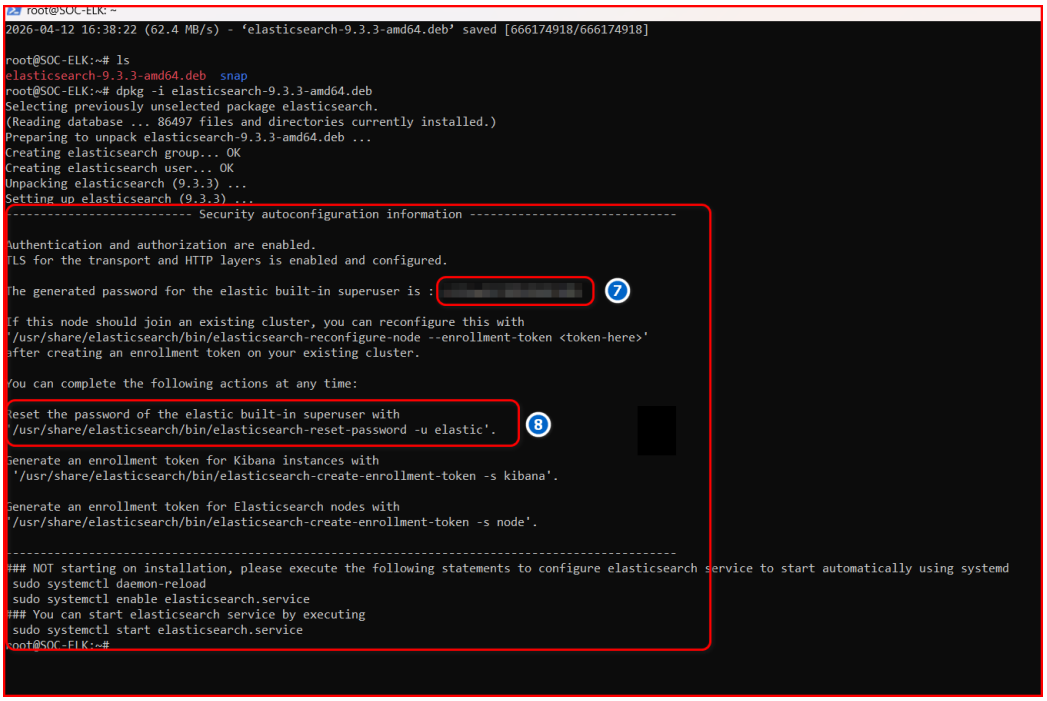



*Figure 17 – Installation output showing generated elastic password*

```
/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic
```

The installer also prints the commands to enable and start the service (we will run them after configuration).

## 19. Explore Configuration Directory

Change into the configuration directory and list the files:

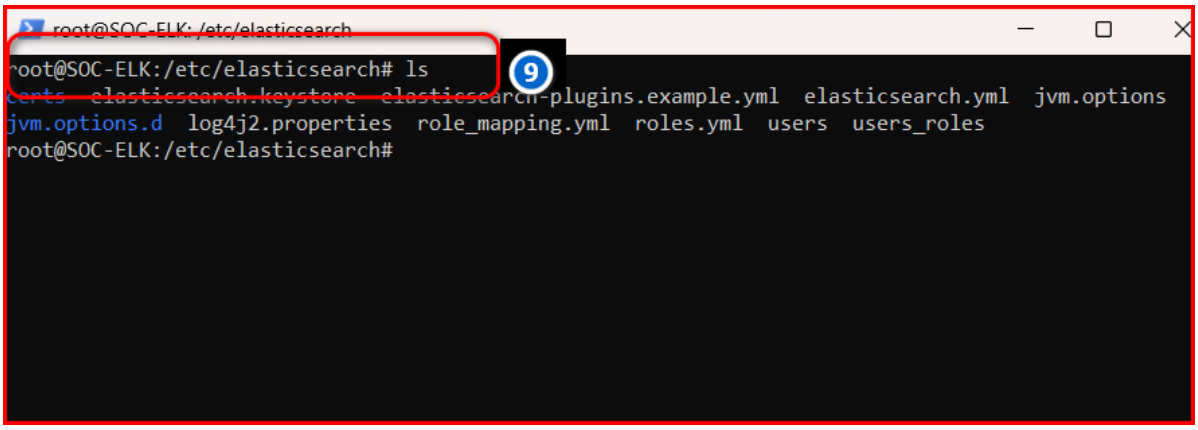



*Figure 18 – Contents of /etc/elasticsearch*

```
cd /etc/elasticsearch && ls
```

Key file: **elasticsearch.yml** – this is where we set network binding.

# 20. Identify Server IP Addresses

Run **ip a** (or **ip addr**) to see both the public interface and the private VPC address (example: 172.31.0.3).

You will use one of these addresses for **network.host**.

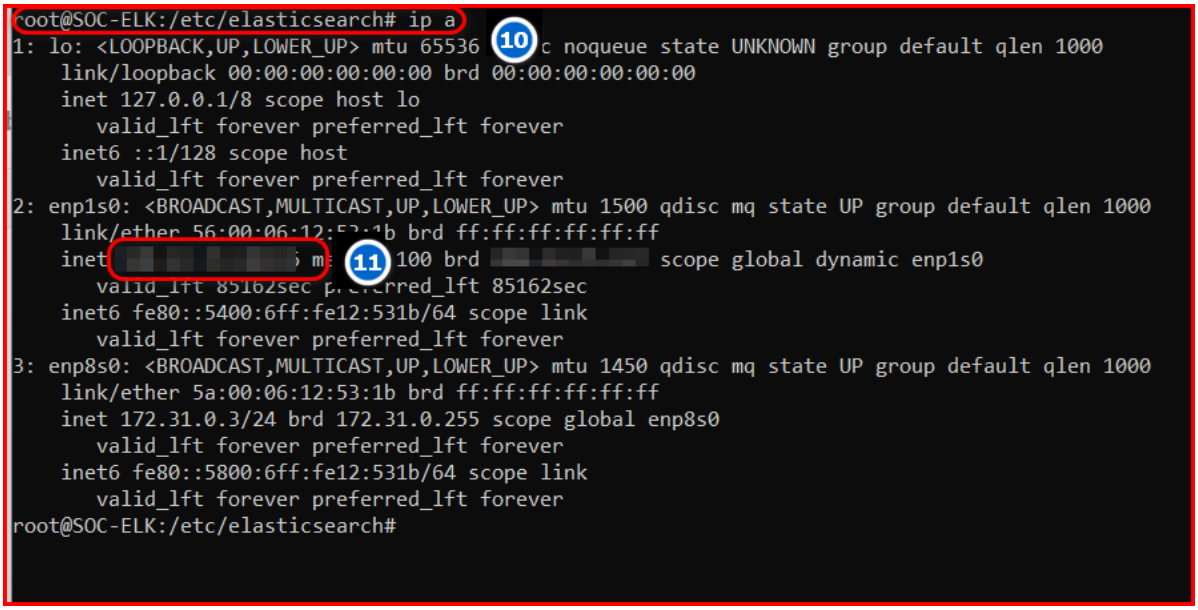



*Figure 19 – ip a showing public and private (172.31.0.3) addresses*

## 21. Edit elasticsearch.yml (network.host)

Open the configuration file:

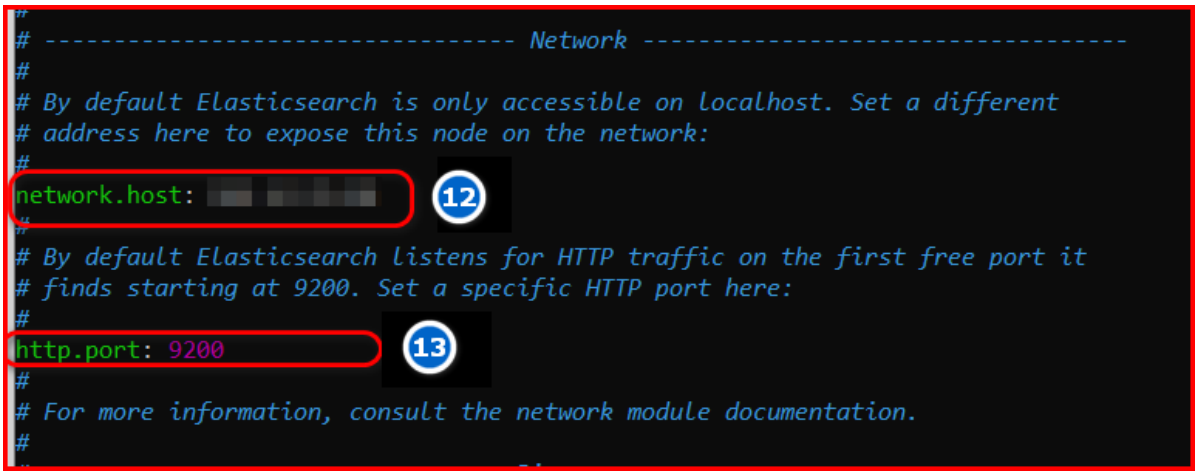



*Figure 20 – network.host and http.port settings*

```
nano /etc/elasticsearch/elasticsearch.yml
```

Locate the Network section and set (replace with your actual IP or use 0.0.0.0 to listen on all interfaces):

```
network.host: <SERVER-IP-OR-0.0.0.0>
http.port: 9200
```

Save and exit (Ctrl+O, Enter, Ctrl+X in nano).

# 22–24. Create & Assign Firewall Group

In the Vultr portal go to the server → **Settings** → **Firewall** → **Manage**.

Create a new Firewall Group (e.g. MYDFIR-SOC) and add inbound rules:

• SSH (port 22) – Source: My IP (or Anywhere for lab use)

• Optionally later: TCP 9200 for Elasticsearch HTTP (restrict source carefully)

Then assign the firewall group to the SOC-ELK instance.

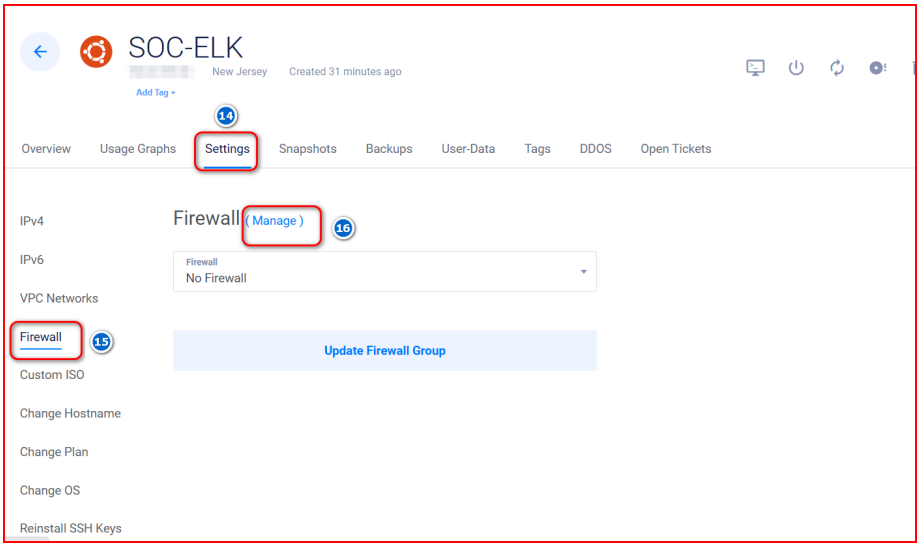



*Figure 21 – Server Settings → Firewall*

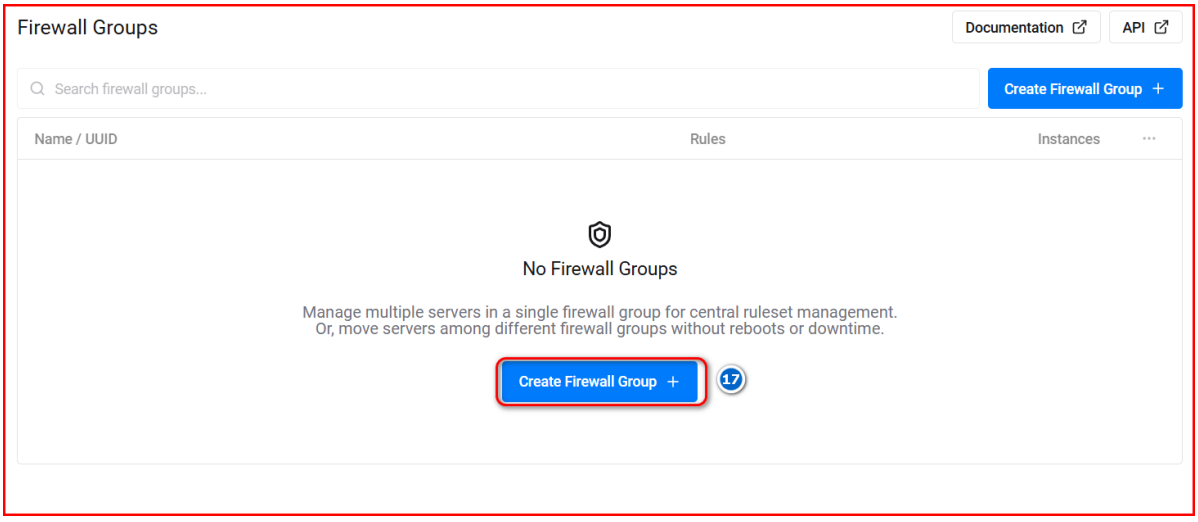



*Figure 22 – Create Firewall Group*

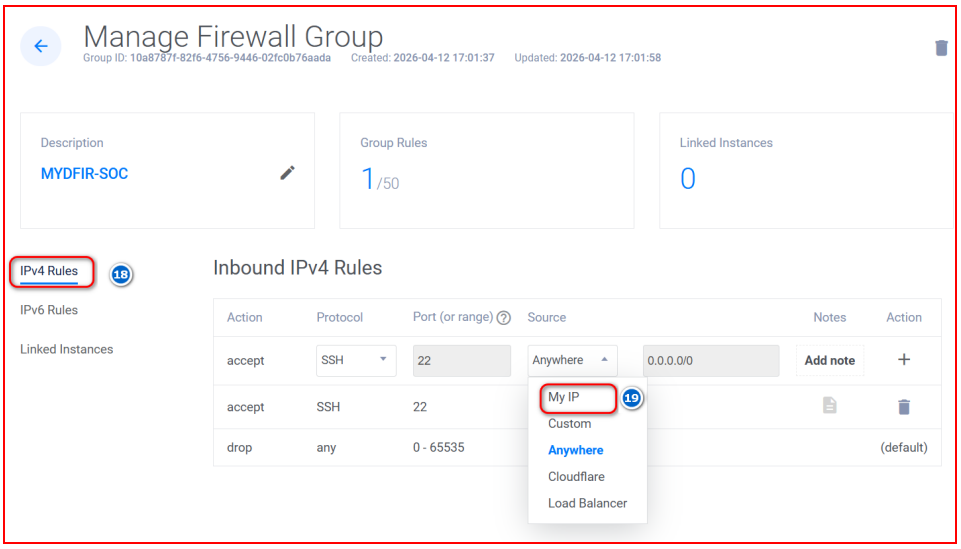



*Figure 23 – IPv4 Rules (SSH)*

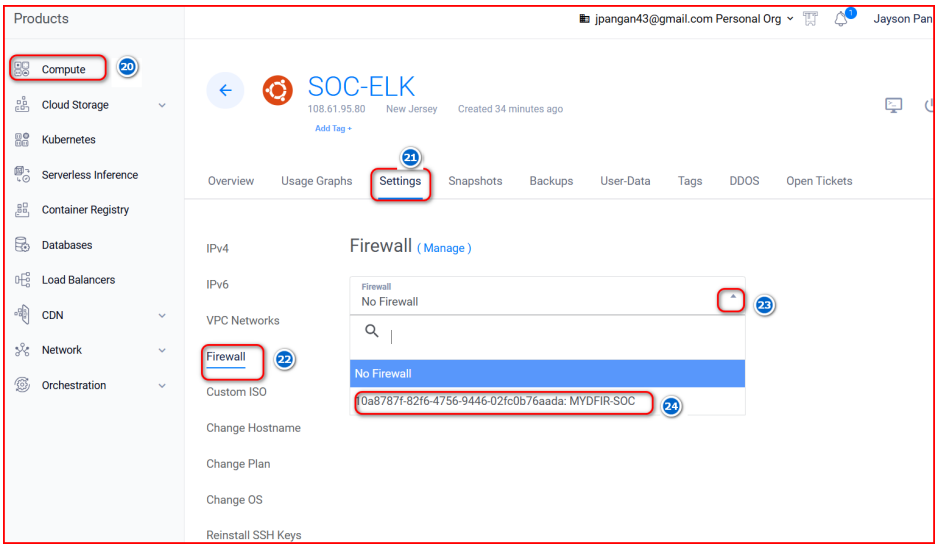



*Figure 24 – Assign firewall group to instance*

## 25. Enable and Start Elasticsearch Service

Reload systemd, enable the service to start on boot, and start it now:

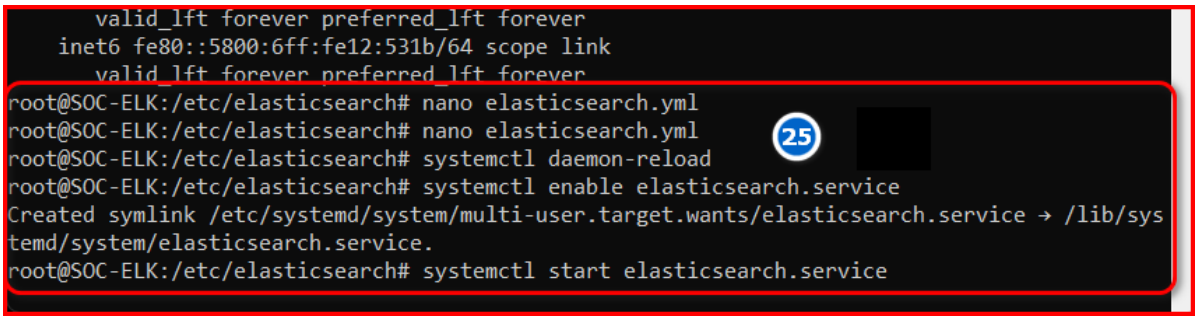



*Figure 25 – systemctl enable & start elasticsearch.service*

```
systemctl daemon-reload
systemctl enable elasticsearch.service
systemctl start elasticsearch.service
```

## 26. Verify Elasticsearch is Running

Check the service status:

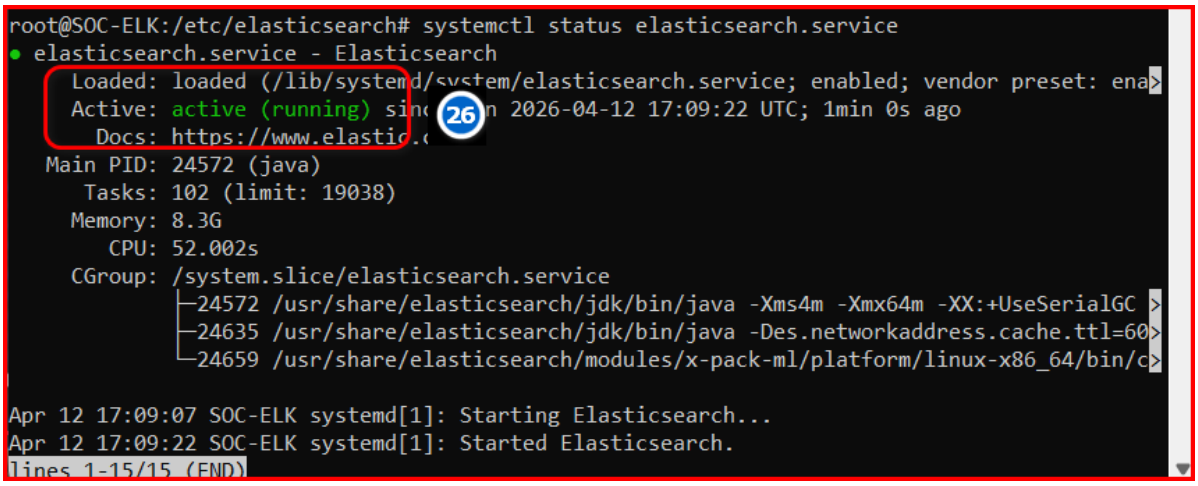



*Figure 26 – systemctl status shows active (running)*

```
systemctl status elasticsearch.service
```

You should see **Active: active (running)**. Elasticsearch is now listening on port 9200.

# 27. Final Server Overview & Next Steps

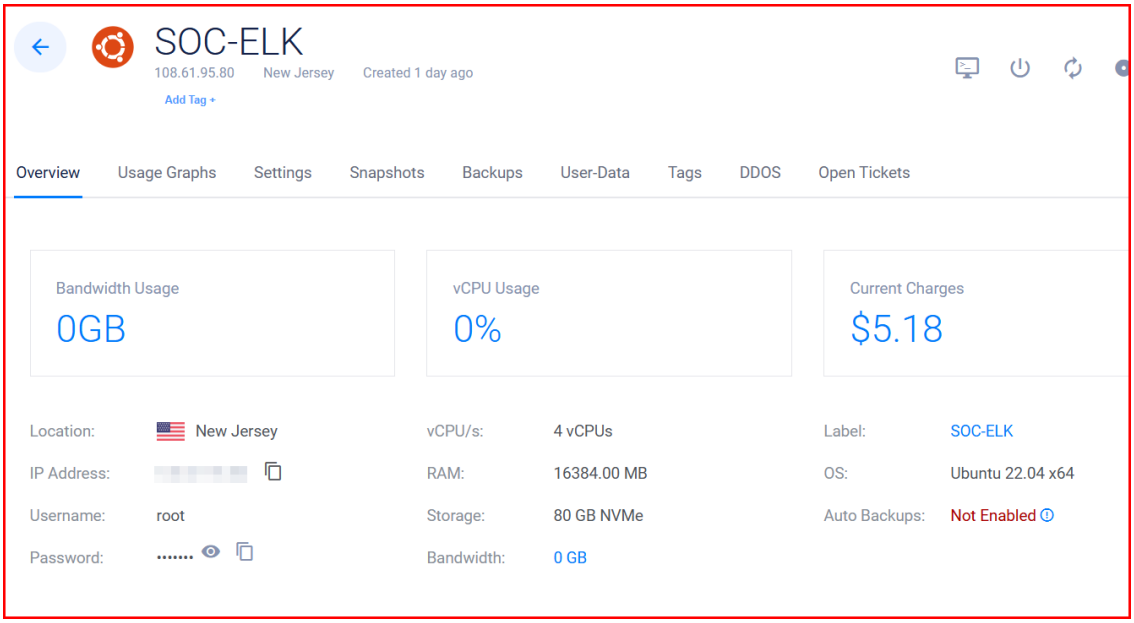



*Figure 27 – SOC-ELK server overview in Vultr portal*

Recommended next actions:

- Test connectivity: `curl -k -u elastic:<password> https://localhost:9200` (or http if TLS not yet required).
- Reset / change the elastic password if you did not save the generated one.
- Install Kibana on the same or another host and enroll it with an enrollment token.
- Configure index templates, ILM policies and any data-ingestion pipelines needed for your SOC use case.
- Restrict firewall rules so that port 9200 is only reachable from trusted sources (never leave it open to 0.0.0.0/0 in production).
- Consider enabling automatic backups or snapshots for the Elasticsearch data directory.

**Security note:** The generated elastic password and any enrollment tokens are sensitive. Store them in a password manager and never commit them to source control.

End of Guide — Elasticsearch Installation on Vultr (SOC-ELK)

*Generated from configuration screenshots • For internal documentation / training use*